

IUTO

Health & Medicine: Predicting the Presence of Heart Disease from Standard Clinical Data

Group Members :

Leni Chabilan / Romain Mechain / **Kylian Riberou**

Project Summary

This project aims to study how standard clinical measurements taken during health check-ups relate to the presence of heart disease, using a public cardiology dataset derived from the UCI/Cleveland Heart Disease database. Each record describes one patient through 13 explanatory variables (age, sex, chest pain type, resting blood pressure, serum cholesterol, resting ECG, maximum heart rate, exercise-induced angina, ST depression, ...) and a binary target indicating whether heart disease is present or absent.

The Problem ?

Cardiovascular diseases remain one of the leading causes of death worldwide, yet risk is driven by many clinical factors acting together rather than any single measurement. Our project asks a question the data can help us actually answer to : from standard basic clinical indicators, can we reliably predict whether a patient has heart disease, and which variables are the most decisive ? Such a data-driven prediction model could help flag at-risk patients early and guide them towards more advanced examinations.

Solution:

After exploring and cleaning the dataset, we analyse the most relevant clinical features and then train and compare four supervised classification models seen in class: Logistic Regression, K-Nearest Neighbors, Decision Tree and Random Forest. Model performance is evaluated with classification metrics (accuracy, precision, recall, F1-score, confusion matrix and ROC/AUC), together with an analysis of which features the models rely on most.

Expected results:

We will test if those clinical indicators are the strongest drivers of the prediction, and report honestly whether the data support this or not.

We anticipate moderate but trustworthy predictive performance rather than a perfect model, and we explicitly acknowledge the dataset's limitations: it originates from a specific North-American population from the 1980s–90s, its demographic origin is not documented, and a single binary label cannot capture the full complexity of a patient's health. The project will follow good data-analysis principles: avoiding data leakage, choosing metrics suited to the task, reporting results honestly, and discussing the real-world risks of misusing medical predictions.